

DB Archival Batch Job

Context

New batch job needs to be implemented for Archival of companies from PSP with an aim to reduce PSP DB size which will result in improved performance and also helps in allowing more companies to cloudify their data as part of Grow Workforce or Historical paychecks send for the workforce enabled companies.

Archive phases:

Phase 1: IOP and QBOE companies

Phase 2: QBDT Inactive companies (no payroll run in last 3 years).

Phase 3: Monthly Schedule Archive company inactive more than 3 years.

Details

1. It will be an on-demand batch job.
2. It will be triggered manually.
3. Archival will only happen on Primary DB. Copy will happen from read replica and purging will happen from main DB (which will then sync with read replica automatically).
4. Partial Archival of a company is not allowed. If there is any issue with archival of a company, we will rollback the entire company records.
5. Once archival is happening successful, we will run this batch job in Automatic mode for QBDT companies (timeline TBD)

JSS parameters

Operation (Mandatory) - Possible operations are

- Archive
- Rollback (this will be same as giving Source as Archival DB and destination as PSP DB)
- Purge
- Archive+Purge

ChunkSize (Optional) - Number of companies to be archived in a batch job run.

SourceSystemCD (Optional) - Default is QBDT, other possible values are QBOE and IOP.

ListPSIDs (Optional) - Default is null. (used in combination with SourceSystemCD. If SourceSystem is provided, and this is null, all companies with given source is fetched)

NamedQuery (Optional) - Default is null.

Number of threads (Optional) - Default (TBD)

Company Retrieval Strategy

List of PSIDs or Query to be executed for retrieving companies to be archived

- For Archival Operation
 - List of companies (for testing purposes) and SourceSystemCD.
 - Query based - For first release (archiving IOP or QBOE companies, JSS parameter SourceSystemCD can be used). For any new strategy to pick companies to be archived, we will write a new query and release.
 - SourceSystemCD (for IOP and QBOE companies) **AND/OR**
 - Inactive Companies (No payroll run since last 3 yrs)
- For Delete Operation
 - List of companies (for testing purposes) - Internally it should verify that a company has been archived before it deletes a given company. This is to ensure avoiding any misuse or inadvertently deleting a company.
 - Query based
 - Archived but not deleted AND
 - SourceSystemCD (for IOP and QBOE companies)

Tracking

We will be creating entries in psp_deleted_record table for company archival and purging to track companies which have been archived and/or deleted.

Other Options

- **Track in Company events table** - We could not track in Company events table, as PSP_COMPANY_EVENT entries for a company will also be archived along with company archival.
- **Add a new column to PSP_COMPANY** - During company archival, this column will also get archived.

Batch Job re-entrancy

If in a given list of companies, archival of one company fails, we will continue with archival of next company.

We will commit a company being archived in Archival DB only once all the tables of the company have been copied. If copying of a table fails in a company, we will rollback the entire operation and this company will be picked up next time. **We will not retry copying a company in this case in a given batch job session.**

Archival Strategy

Given a list of companies to be archived, we have multiple options available for archival:

Archival per company (Single threaded)

One company per thread but multiple companies in parallel (Multithreading)

Chunk of companies queued in multiple threads (Multithreading)

Other design decisions

If a company/employee record has been deleted as requested by customer using DG Service, does it actually delete it from DB? Do we need to consider this in our company retrieval queries? - Today when a DELETE request is raised for a company, the company is not actually deleted, it is only marked for deletion in PSP. **Yes, we will consider this company for archival and deletion from source DB.**

Alerts